Tide retrospective:
Mixed odd-power 2D moment vanishing (V1)

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Abstract

This retrospective records a small parity-completion pair on top of the quartic–sextic 2D potential machinery established by Tide A2 and used by Tide T1. For $L(x, y) = x^4/24 + y^6/720$ at $t > 0$, the two theorems state that any joint moment $\langle x^a y^b \rangle_{2D,t}$ vanishes when at least one of a, b is odd, with no case-split: by leaving the non-odd index as a free natural number rather than forcing it to $2k$, the pair of theorems together covers the odd-odd case as well. Each proof is a single `rw` block (factor theorem + 1D odd-vanishing) closed by `simp`; the file is 80 lines including docstrings. Single session, build green on first compile after the integrability and 1D lemmas were located. Zero `sorry`, zero `axiom`, zero `native_decide`.

1 Setting

The grammar-paper precursor sequence and the 2D quartic–sextic sub-thread of Tide A2 left T1 as the first joint-moment theorem on the new 2D potential: the even-power closed form $\langle x^{2j} y^{2k} \rangle_{2D,t}$. T1's retrospective explicitly flagged the parity-completion pair as a deferred follow-up (T1's "Candidate C"): the mirror odd-vanishing theorems needed to give a full picture of mixed-monomial moments under the separable measure. The 1D building blocks[1] were already in the seabed, both proven unconditionally in t via odd-symmetry of the integrand against the even-symmetric exponential factor. A2's factor theorem[2] provides the 2D-to-1D reduction. With all three pieces in place, V1 is the minimal step that finishes the parity story.

2 The thing formalised

For the 2D potential $L(x, y) = x^4/24 + y^6/720$ at temperature $t > 0$:

- For all $j, n \in \mathbb{N}$, $\langle x^{2j+1} y^n \rangle_{2D,t} = 0$.
- For all $m, k \in \mathbb{N}$, $\langle x^m y^{2k+1} \rangle_{2D,t} = 0$.

The two theorems together cover every $\langle x^a y^b \rangle_{2D,t}$ with at least one of a, b odd, including the odd-odd case. Lean signatures:

```
theorem gibbsExpectation_quarticSextic_odd_pow_pow_eq_zero
    (j n : ℕ) {t : ℝ} (ht : 0 < t) :
```

[1] Specifically `Laplace.OneD.quartic_expected_value_odd` and `Laplace.OneD.sextic_expected_value_odd`.
[2] `Laplace.TwoD.gibbsExpectation_quarticSextic_factor`, in `Laplace/TwoD/QuarticSextic.lean`.

```
    gibbsExpectation
        (addSeparable Laplace.OneD.quarticPotential
                      Laplace.OneD.sexticPotential) t
        (fun z : ℝ × ℝ => z.1 ^ (2 * j + 1) * z.2 ^ n) = 0

theorem gibbsExpectation_quarticSextic_pow_odd_pow_eq_zero
    (m k : ℕ) {t : ℝ} (ht : 0 < t) :
    gibbsExpectation
        (addSeparable Laplace.OneD.quarticPotential
                      Laplace.OneD.sexticPotential) t
        (fun z : ℝ × ℝ => z.1 ^ m * z.2 ^ (2 * k + 1)) = 0
```

Both live in the `Laplace.TwoD` namespace, conforming to T1's even-case naming.[3] The new file is `Laplace/TwoD/QuarticSexticOddMoments.lean`, sibling to T1's `QuarticSexticMoments.lean`, and is added to the project root `Laplace.lean`.

3 Proof strategy

For the odd-x side: apply the factor theorem with $f = x \mapsto x^{2j+1}$ and $g = y \mapsto y^n$. The 2D expectation factors as a product of the 1D quartic expectation and the 1D sextic expectation. The first factor is zero by `quartic_expected_value_odd`. The product is zero, and `simp` closes the residual `0 * _` via `zero_mul`. The odd-y side is the mirror image: the *second* factor is zero by `sextic_expected_value_odd`, `simp` closes via `mul_zero`.

The integrability hypotheses fed to the factor theorem are the canonical `_integrable_pow_pot` witnesses[4], the same shape T1 used. They unify with the factor theorem's expected form definitionally; no `change`-tactic bridging needed.

4 Roadblocks and resolutions

There were no roadblocks. The proof template was correct on first write; `lake build` returned green on the first compile. The deliberation surfaced one strict-improvement over the user's draft (see §5), so the implementation step ran on the deliberation's revised plan rather than the user's initial sketch.

5 What was Mathlib, what was new

From Mathlib. Nothing direct in this tide. The factor theorem and the 1D odd-vanishing theorems are Laplace-side; the only Mathlib-side dependency is `zero_mul` / `mul_zero` inside `simp`.

From the seabed. Three pieces did the work: A2's factorisation engine, the 1D quartic and sextic odd-vanishing theorems (both unconditional in t), and the canonical integrability lemmas. Names recorded in the footnotes of the §*Setting* and §*Proof strategy* sections above.

[3] T1's headline theorem is `gibbsExpectation_quarticSextic_pow_pow_eq`; the new pair mirrors that name with an `odd`-infix and `_eq_zero` suffix.

[4] Namely `Laplace.OneD.quartic_integrable_pow_pot` and `Laplace.OneD.sextic_integrable_pow_pot`.

New here. Two parity-completion theorems, 80 lines total (including docstrings). The architectural contribution is the choice to leave the non-odd index as a free n (resp. m) rather than forcing it to $2k$ (resp. $2j$): this is what makes the pair cover the odd-odd case without a case split or an `Odd`-witness predicate. The contribution came from GPT-5.5 Pro at Step 2; see the Lessons section.

6 Lessons

Free-the-other-index is a strict improvement on forced-even. The user's initial draft had the two theorems keyed on $(2j+1, 2k)$ and $(2j, 2k+1)$, mirroring T1's even-case $(2j, 2k)$ structure. GPT-5.5 Pro pointed out this misses the odd-odd case and recommended $(2j+1, n)$ and $(m, 2k+1)$: same proof size, no case split, total coverage. The pattern generalises: *when proving a vanishing identity that factors through a single factor's vanishing, leave the other factor's parameter free rather than mirror the parametrisation of the matching positive identity.* The matching positive identity (T1) needs both factors to have closed forms, hence the symmetric $(2j, 2k)$; the vanishing identity (V1) only needs one factor to vanish, so the symmetry breaks and the freer parametrisation costs nothing.

`simp` over `ring` for residual `0 * _` goals. GPT also suggested `simp` in place of the draft's `ring` at the bottom of each proof. After the two rewrites, the goal is literally `0 * _ = 0` or `_ * 0 = 0`; `simp` handles this via `zero_mul` / `mul_zero` (and `simp` is at this point cheaper than `ring`, which would also work but would go through a normalisation pass). Worth noting as a tactic preference for this class of small wrap-up goal.

The single-round consensus. Both Claude and GPT-5.5 Pro agreed on Candidate A′ in one round. The vote, the integrability unification, the bottom-tactic choice, and the rejection of the generic `addSeparable` abstraction ("one tide too far") all landed in the same consult. Pattern: small tides off a recent retrospective's flagged Candidate often have one canonically-right shape that a single GPT round nails. The cost of the consult was recouped by the strict improvement on the user's draft.

7 Follow-ups

- *Generic `addSeparable` odd-vanishing abstraction.* For an arbitrary additively separable 2D potential $V(x, y) = V_1(x) + V_2(y)$ where V_1 is even-symmetric, any monomial moment with an odd x-exponent vanishes. The 1D abstract prerequisite is "$\int x^{2j+1} e^{-tV(x)}\, dx = 0$ for V even-symmetric" — which the seabed does not yet have as a generic lemma (only the quartic and sextic specialisations). When a third or fourth even-symmetric 1D potential lands (e.g. a pure k-th-power potential beyond $k = 4, 6$), promote the generic abstraction and rewrite the quartic and sextic odd-vanishings as specialisations.

- *Higher-power named consumers.* If a downstream consumer ever wants a specific named family $\langle x^{2j+1} y^{2k} \rangle_{2D,t} = 0$ (rather than the generic $(2j + 1, n)$ form), it is a one-line corollary of V1's first theorem at the call site; no follow-up tide needed.

- *`\leanref`-tag the primer's 2D examples* once the primer's TeX has statements for the 2D parity-completion identities. Pin to this tide's commit. Pairs with the existing K2 / U3 / L5 paper-tagging backlog.

Acknowledgements

GPT-5.5 Pro's deliberation contributed three load-bearing pieces of guidance in a single round: the parametrisation strengthening from $(2j+1, 2k)$ to $(2j+1, n)$ that closes the odd-odd hole the draft missed; `simp` as the bottom tactic in place of `ring`; and the `_eq_zero` naming suffix. The proof landed single-attempt with no thrashing-rescue consult; the full GPT response is preserved alongside the tide log in the primer project's tide-log folder.

This Tide branched off the post-L3-merge `main`, the same `main` that gathered the seven-tide backlog merged earlier on 2026-05-18 (the day's bookkeeping cleanup). The fresh `main` contains A2, T1, U2, R1, Q1, L2, L3 inline rather than as detached tide branches.